Factor Analysis for Road Accident Severity Classification Using Machine Techniques

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***Abstract*—This paper aims to develop a model for classifying road accident severity into three levels: Severe (KA), Minor (BC), and No Injury (O). The study used the Traffic Accidents dataset from Kaggle, consisting of 209,306 rows and 18 variables. After data cleaning, 171,519 rows and 17 variables remained. The dataset was then split into 80% training data and 20% testing data. Categorical variables were transformed using One-Hot Encoding, class imbalance was addressed through undersampling, and feature selection was performed using the Chi-square test at a significance level of p-value < 0.05. Subsequently, predictive models were developed using Random Forest, LightGBM, and CatBoost, and their outputs were combined through a Soft Voting Ensemble approach. Hyperparameter optimization was conducted using Grid Search Cross-Validation, and model performance was evaluated using Accuracy, Precision, Recall, and F1-Score. The experimental results showed that Random Forest achieved the highest F1-Macro score of 0.5433, while CatBoost demonstrated the best performance in detecting severe accidents, with a Recall score of 0.5269. Nevertheless, challenges remain in improving the model's effectiveness for classifying the KA class, highlighting that class imbalance continues to be a major issue**

Keywords—classification, machine learning, road accident, traffic accident

I. Introduction

Road traffic accidents claim millions of lives each year. According to the World Health Organization (WHO, 2018), approximately 1.35 million people die annually from road accidents, making it a serious global public health issue. In Thailand, the impact extends beyond physical injuries and property damage, as it also directly affects the economy because most victims are within the working-age population. Therefore, understanding the factors that indicate accident severity is highly valuable, as such information can help policymakers design more targeted and effective prevention strategies [1].

At present, AI and Machine Learning technologies have enabled accident data analysis to become far more detailed than before. Multiple dimensions of data—including weather conditions, time periods, road characteristics, and visibility levels—can now be analyzed simultaneously.

This study applies Machine Learning techniques to develop a model capable of identifying the severity level of each accident case. The research utilizes the Traffic Accidents dataset created by Oktay Ördekçi and published on Kaggle [2], in order to demonstrate how classification results from such analysis can support policy-oriented decision-making.

II. Literature Review

A. Chi-Square Test

The Chi-Square Test is a statistical tool used to determine whether two categorical variables are truly associated with each other. The main concept involves comparing the observed frequencies with the expected frequencies that would occur if the two variables were completely independent. If the difference between these two values exceeds a predefined threshold, it can be concluded that the variables are significantly related [3].

In this paper, the Chi-Square Test was employed as a feature selection filter. Only variables with a p-value lower than 0.05 were retained, allowing irrelevant variables that did not contribute to model performance to be removed before the model development process began.

B. Encoding

Most Machine Learning (ML) algorithms can only process numerical data. Therefore, textual or categorical data must first be transformed into numerical representations. This study employed the One-Hot Encoding technique, which converts each category into a separate column and assigns binary values of 0 and 1 to indicate the category membership of each data instance.

The advantage of this method is that it prevents the model from incorrectly interpreting categorical values as having an ordinal relationship. This technique is particularly well suited for tree-based algorithms and other computational methods commonly used in classification tasks [4].

C. Random Forest (RF)

Random Forest is based on the concept that combining the outputs of multiple models can achieve higher accuracy than relying on a single model. This computational approach constructs a large number of decision trees, where each tree is trained on a randomly sampled subset of the dataset using bootstrap sampling with replacement. At each split node, only a random subset of features is considered for decision-making.

The diversity generated by this mechanism causes each decision tree to have different characteristics. When their predictions are aggregated through majority voting, the errors

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from individual trees tend to offset one another, resulting in a more stable and accurate overall prediction [5].

Wang and colleagues (2024) applied Random Forest to predict traffic accident severity levels. Their findings demonstrated that this computational approach was highly effective in handling complex relationships within the data, particularly environmental and infrastructural factors that significantly influenced prediction performance [6].

D. LightGBM (Light Gradient Boosting Machine)

LightGBM is a Gradient Boosting computational approach developed by Microsoft with the primary objective of improving speed and scalability for large-scale datasets. One of the key differences between LightGBM and XGBoost lies in its leaf-wise tree growth strategy. In each iteration, LightGBM expands the leaf that provides the greatest reduction in the loss function, rather than expanding all leaves level by level as performed in XGBoost.

In addition, LightGBM utilizes a Histogram-based Algorithm for determining feature split points, significantly reducing both computation time and memory consumption. As a result, it is particularly suitable for tasks involving large numbers of variables and requiring high practical performance [7].

Li, Xu, and Liu (2022) further enhanced this approach by integrating LightGBM with the Tree-structured Parzen Estimator (TPE) hyperparameter optimization technique. Their study demonstrated that the resulting model outperformed both Random Forest and XGBoost across multiple evaluation metrics in accident severity classification tasks. Furthermore, the model clearly identified influential variables such as weather conditions, road characteristics, and the number of vehicles involved. These findings highlight LightGBM's strengths in terms of accuracy, computational efficiency, and model interpretability [8].

E. CatBoost

Prokhorenkova and colleagues (2018) introduced CatBoost as a Gradient Boosting approach specifically designed to address the Prediction Shift problem, which refers to statistical bias arising when the same dataset is used for both training and evaluating a model. They proposed two key innovations: Ordered Boosting, which applies data ordering principles to prevent target leakage, and Ordered Target Statistics, which transforms categorical features into numerical values without causing information leakage.

In addition, CatBoost employs Oblivious Decision Trees as its base predictors. These symmetric decision trees help reduce overfitting while also improving inference speed. Experimental comparisons against XGBoost and LightGBM across nine benchmark datasets showed that CatBoost outperformed both models on all datasets, particularly those containing large numbers of categorical features [9].

Roudnitski (2024) combined five Machine Learning models—Random Forest, XGBoost, AdaBoost, LightGBM, and CatBoost—to predict accident severity using data from New South Wales, Australia, collected between 2018 and 2022. The dataset contained 57,240 records and included information related to road conditions, collision patterns, vehicle types, and environmental factors.

Because fatal accidents accounted for only 2.5% of the dataset, the SMOTE technique was applied to address class imbalance before combining model predictions using Soft Voting. The resulting ensemble achieved a ROC-AUC score of 0.68. The most influential factors were identified as the primary vehicle type and the first collision pattern. The study also found that severe accidents were more likely to occur on Saturdays on rural roads with speed limits of 100 km/h [10].

F. Differences Between This Study and Previous Research

Although previous studies have demonstrated the potential of tree-based models in classifying road accident severity, most research has focused either on single models or on ensemble approaches that combine multiple models with highly similar learning mechanisms.

This study therefore proposes a comparative framework between single models and an ensemble of three models—Random Forest, LightGBM, and CatBoost—selected based on the diversity of their learning mechanisms, combining both Bagging and Boosting approaches. The objective is to classify accident severity into three levels according to the KABCO standard.

G. Rationale for Selecting Random Forest, LightGBM, and CatBoost

This study selected Random Forest, LightGBM, and CatBoost for developing the accident severity classification model for two main reasons.

First, all three models belong to the Tree-based Ensemble family, which is fundamentally built on decision tree structures. These models are highly effective in handling categorical and mixed-type data. This aligns well with the characteristics of the Traffic Accidents dataset, which contains many categorical variables such as weather conditions, road characteristics, and accident types.

Second, the three models employ different learning mechanisms. Random Forest is based on the Bagging approach, where decision trees are trained in parallel and their outputs are combined using majority voting. In contrast, LightGBM and CatBoost are based on Boosting methods, where trees are built sequentially and each model learns from the errors of its predecessors. This diversity in learning strategies leads to uncorrelated errors among the models, which is a key condition for Ensemble Soft Voting to effectively reduce overall prediction errors and improve the stability and accuracy of the final results.

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H. Grid Search Cross-Validation

Grid Search Cross-Validation is a hyperparameter tuning technique that defines a set of possible values for each parameter in the form of a grid. It then exhaustively tests all possible combinations and selects the set of parameters that yields the best performance according to a predefined evaluation metric [13].

To evaluate each parameter combination, K-Fold Cross-Validation is used. This is an evaluation method designed to reduce bias from a single train-test split. The process involves dividing the training set into K equal parts, then training and evaluating the model K times. In each iteration, K−1 folds are used for training, while the remaining fold is used for validation. This ensures that every data point is used for both training and validation exactly once. The final performance of each parameter set is calculated as the average score across all K iterations [14].

In this study, K is set to 5, which is widely accepted in machine learning research as it provides a good balance between evaluation reliability and computational cost [14]. Setting K = 5 means that the training set is split into 5 parts. In each cross-validation round, 4 parts (80% of the training set) are used for training, while the remaining 1 part (20%) is used for validation.

In addition, this study employs Stratified K-Fold Cross-Validation, which is a variation of K-Fold that preserves the proportion of each class across all folds to match the original distribution of the training set. This is particularly important in this study because the dataset is imbalanced. If standard K-Fold were used, some folds might contain very few or even no samples of the minority class KA, which would reduce the reliability of the evaluation results.

III. METHODOLOGY

A. Process overview

The framework of this simulation aims to develop a model capable of classifying the severity level of each road traffic accident. The study uses Random Forest, LightGBM, CatBoost, and an Ensemble approach. The overall workflow is illustrated in Figure 1. The details of each step are as follows:

1) Import Data

The dataset used in this study is the Traffic Accidents Dataset by Oktay Ördekçi, obtained from the Kaggle platform [2]. It consists of 209,306 rows and 18 variables, covering key information such as the date and time of the accident, road type and surface conditions, weather conditions, number of injuries, and the severity level of the accident, which serves as the target variable.

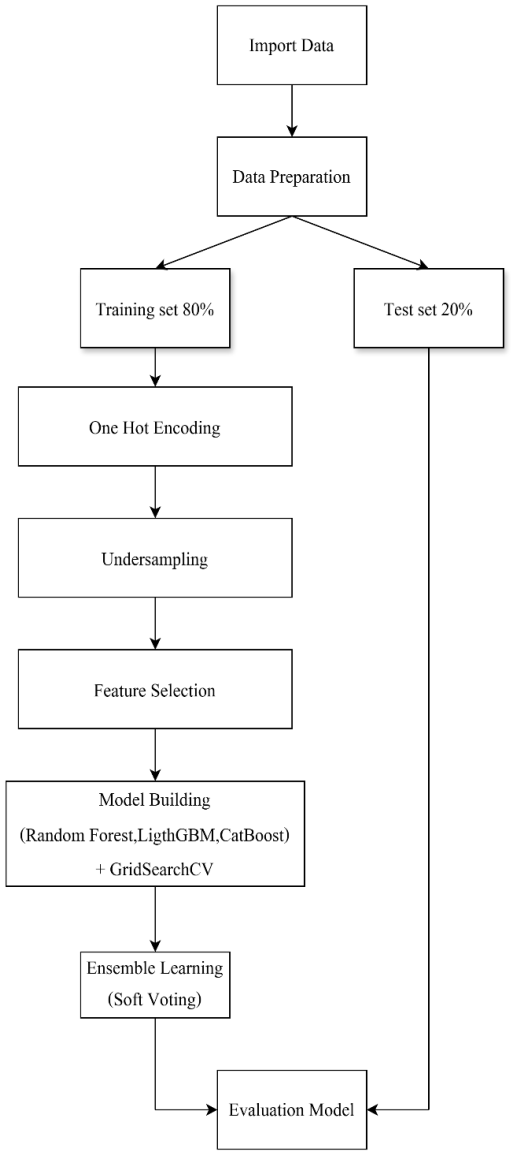


Fig. 1. Overall Operational Process Diagram

For initial processing, the data was imported using Python and handled with the main libraries including pandas, NumPy, and scikit-learn, in order to prepare the dataset for statistical analysis and subsequent model development.

2) Data Preparation

This step involves cleaning and preparing the data for analysis. The severity levels of traffic accidents were consolidated from the original five categories (FATAL, INCAPACITATING INJURY, NONINCAPACITATING INJURY, REPORTED NOT EVIDENT, and NO

INDICATION OF INJURY) into three levels, based on the KABCO standard classification framework for road traffic injury severity [11]. The grouping is defined as follows:

- **Severe (KA):** Includes fatal cases and severe injuries, namely FATAL and INCAPACITATING INJURY.
- **Minor (BC):** Includes minor injuries or cases where injury is reported but not clearly evident, namely NONINCAPACITATING INJURY and REPORTED NOT EVIDENT.
- **No Injury (O):** Cases with no indication of injury, namely NO INDICATION OF INJURY.

3) Data Splitting

After data preparation, the dataset was divided into two parts: an 80% training set for model training and a 20% test set for model evaluation. The split was performed using the train_test_split() function from scikit-learn, with Stratified Sampling applied to ensure that the class distribution in both subsets remained similar.

4) Encoding

In this step, categorical variables were converted into numerical format using One-Hot Encoding for all independent variables. For example, the variable "Weather Condition," which contains five categories, is transformed into five new columns. Each column contains binary values (0 or 1) indicating whether a given record belongs to that category.

5) Undersampling

The dataset used in this study has a significant class imbalance problem, where the No Injury class contains many times more samples than the Severe class. If the model is trained directly on such data, it is likely to be biased toward predicting the majority class while ignoring minority classes. Therefore, the researcher applied Undersampling to address this issue.

This method involves randomly reducing samples from the majority class until its size matches that of the minority class. This process is applied only to the training set and not to the test set. If the test set were also modified, the evaluation results would not reflect real-world performance.

6) Feature Selection

Redundant variables were removed to improve model efficiency and learning performance. The researcher applied the Chi-Square test to filter independent variables, retaining only those with a p-value lower than 0.05 for use in the next stage of model development.

7) Model Building

Three classification models were developed:

- **Random Forest (RF)** — Constructs multiple decision trees from randomly sampled subsets of the data and aggregates their predictions through majority voting. The final prediction is determined by the class with the most votes, resulting in more stable and less variable outputs.
- **LightGBM** — Based on the same boosting principle as XGBoost but optimized for faster computation using a Leaf-wise tree growth strategy, where only the leaf that most reduces the error is expanded at each step.
- **CatBoost** — A Gradient Boosting technique developed by Prokhorenkova et al. (2018). It can directly handle categorical variables without prior encoding and introduces Ordered Boosting, which orders data before training to reduce Prediction Shift and Target Leakage issues commonly found in traditional Gradient Boosting methods. Its base learner is Oblivious Decision Trees, which have a symmetric structure that helps control overfitting and improves computational efficiency.

8) Hyperparameter Tuning

To optimize model performance, hyperparameter tuning was conducted using GridSearchCV. Stratified K-Fold Cross-Validation with K = 5 was applied for each model. During the tuning process, the evaluation metric used was F1-macro, which assigns equal weight to all classes. This helps reflect model performance in situations where the dataset is imbalanced across classes.

9) Ensemble Learning

The three models—Random Forest, LightGBM, and CatBoost—were combined using the Soft Voting Ensemble method. Zhou (2012) explained that combining multiple models with different strengths can reduce prediction errors and improve stability compared to using a single model.

Soft Voting differs from Hard Voting in that it does not simply count the majority class predictions. Instead, it averages the predicted probabilities from each model and selects the class with the highest average probability. This approach therefore makes fuller use of the confidence information provided by the models [12].

The reasons for choosing Soft Voting are threefold. First, it uses probability values rather than simple voting counts, allowing it to better reflect prediction confidence. Second, Kuncheva (2014) noted that Soft Voting performs better than Hard Voting in imbalanced classification problems, which is relevant to this study where the KA class is significantly underrepresented [15]. Third, Niculescu-Mizil and Caruana (2005) found that Soft Voting performs well when the combined models produce well-calibrated probability estimates, which is the case for Random Forest, LightGBM, and CatBoost [16].

The final prediction using Soft Voting is computed according to the equation proposed by Zhou as follows:

$$\hat{y} = argmax_c \sum_{i=1}^{3} w_i \cdot P_i(c) \quad (1)$$

Here, Pi(c)P_i(c)Pi(c) represents the probability predicted by model iii that a given sample belongs to severity class ccc, and wi=13w_i = \frac{1}{3}wi=31 for all models, as equal weights are assigned so that each model contributes equally to the final result.

All three models were trained on the same training set and evaluated on the same test set. The F1-macro score was used as the primary evaluation metric to compare the performance between individual models and the ensemble approach.

10) Model Evaluation

After the hyperparameters were tuned and the most suitable models were obtained, they were evaluated on the test set using the following metrics:

- **Accuracy:** Measures the overall correctness of the model's predictions.
- **Precision, Recall, and F1-Score (Macro Average):** Evaluate the model's ability to classify each class, treating all classes equally.
- **Confusion Matrix:** Shows the number of correct and incorrect predictions for each class in detail.

The results from all four models (RF, LightGBM, CatBoost, and the Ensemble) were then compared to determine which model is most suitable for classifying traffic accident severity.

IV. Experimental And Results

A. Data Preparation

The data used in this study was obtained from the "Traffic Accidents" dataset on Kaggle, consisting of 209,306 rows and 18 variables. After data cleaning, records containing unknown values ("UNKNOWN") were removed, resulting in a final dataset of 171,519 rows and 17 variables for analysis. The severity levels of accidents were also consolidated from five categories into three levels based on the KABCO classification framework as follows:

- **Severe (KA):** Includes FATAL and INCAPACITATING INJURY, totaling 5,941 rows.
- **Minor (BC):** Includes NONINCAPACITATING INJURY and REPORTED NOT EVIDENT, totaling 40,049 rows.
- **No Injury (O):** Includes NO INDICATION OF INJURY, totaling 125,529 rows.

B. Data Splitting

The dataset was split using the train_test_split() function from the Scikit-learn library into two parts as follows: an 80% training set containing 137,215 rows and a 20% test set containing 34,304 rows. Stratified Sampling was applied to preserve the proportion of each class in the target variable across both the training and test sets. The detailed distribution is shown in Table I.

TABLE I. Number of samples in each class for the Training set and Test set

Class	Training set (rows)	Test set (rows)
Severe (K+A)	4,753	1,188
Minor (B+C)	32,039	8,010
No Injury (O)	100,423	25,106
Total	137,215	34,304

C. Encoding

One-Hot Encoding was applied to all categorical independent variables in order to convert the training data into a numerical format that can be processed by the models. Each variable was expanded into multiple binary features based on the number of categories it contained, where values of 0 or 1 indicate the presence of each category.

After the One-Hot Encoding process, the dataset contained a total of 140 features, making it ready for model training in the next stage.

D. Undersampling

Due to class imbalance in the dataset, Undersampling was applied to balance the training data before model training. This was done by randomly reducing the number of samples in the majority classes to match the size of the minority class. The results of the Undersampling process are shown in Table II.

TABLE II. Number of samples in each class before and after Undersampling

Class	Before Undersampling (rows)	After Undersampling (rows)
Severe (K+A)	4,753	4,753
Minor (B+C)	32,039	4,753
No Injury (O)	100,423	4,753
Total	137,215	14,259

From Table II, it can be observed that after applying the Undersampling technique, the Training set became fully balanced, with an equal number of samples in each class at 4,753 rows per class, resulting in a total of 14,259 rows. The Undersampling process was applied only to the training set, while the test set retained its original distribution to ensure that the evaluation reflects real-world conditions.

E. Feature Selection

After applying the Chi-Square filter with a significance level of p-value < 0.05, 51 variables out of the original 140 features were selected. These 51 variables were then used across all three models.

When analyzing feature importance from the trained models, it was found that each learning algorithm produced different rankings of feature importance, as shown in Table III.

TABLE III. Top 3 Most Important Features of Each Model

Class	Rank	Feature
RF	1	cat__crash_type_INJURY AND / OR TOW DUE TO CRASH
	2	cat__crash_type_NO INJURY / DRIVE AWAY
	3	cat__first_crash_type_PEDESTRIAN
LightGBM	1	num__crash_hour
	2	num__num_units
	3	cat__crash_type_INJURY AND / OR TOW DUE TO CRASH
CatBoost	1	cat__crash_type_INJURY AND / OR TOW DUE TO CRASH
	2	cat__crash_type_NO INJURY / DRIVE AWAY
	3	cat__first_crash_type_PEDESTRIAN

From Table III, it can be observed that Random Forest and CatBoost produce consistent rankings of feature importance. Both models identify the accident type involving injury and/or towing due to crash (cat__crash_type INJURY AND/OR TOW DUE TO CRASH) as the most important feature.

In contrast, LightGBM places greater importance on numerical variables, ranking crash time (num__crash_hour) and the number of vehicles involved (num__num_units) as the top features.

These differences reflect the distinct learning characteristics of each computational approach.

F. Model Building & Tuning

The models were trained and their hyperparameters were optimized using Grid Search Cross-Validation (5-fold) to obtain the most suitable parameter settings. The F1-macro score was used as the evaluation metric. The best-performing hyperparameters (Best Parameters) are presented in Table IV.

TABLE IV. HYPERPARAMETER TUNING RESULTS OF THE MODELS USING GRID SEARCH CROSS VALIDATION

Model	Best Parameters	F1-Macro
RF	{'max_depth':10, 'min_samples_leaf':2, 'min_samples_split':5, 'n_estimators':300}	0.5393
LightGBM	{'colsample_bytree':1.0, 'learning_rate':0.05, 'max_depth':5, 'min_child_samples':10, 'n_estimators':200, 'num_leaves':15, 'subsample':0.6}	0.5343
CatBoost	{'depth':6, 'iterations':1000, 'l2_leaf_reg':7, 'learning_rate':0.03}	0.5321

G. Ensemble Learning

In this study, the three models—Random Forest, LightGBM, and CatBoost—were combined using the Soft Voting Ensemble method. The final prediction was obtained by averaging the predicted probabilities from each model with equal weights (w = 1/3), and selecting the class with the highest combined probability as the final output.

H. Model Evaluation

- *Random Forest*

The performance evaluation results of the Random Forest model on the test set are shown in Table V.

TABLE V. Accuracy, PRECISION, Recall, and F1-Score of Random Forest

Class	precision	recall	f1-score
KA (Severe)	0.1282	0.4966	0.2038
BC (Minor)	0.4946	0.6783	0.5721
O (No Injury)	0.9999	0.7455	0.8541
Accuracy			0.7212
Macro avg	0.5409	0.6401	0.5433
Weighted avg	0.8517	0.7212	0.7658

From Table V, it can be observed that the Random Forest model achieved an Accuracy of 0.7212 and an F1-macro score of 0.5433. When evaluated by class, the model performed well in classifying the O (No Injury) class, with an F1-score of 0.8541. However, it showed relatively poor performance in identifying the KA (Severe) class, with an F1-score of only 0.2038.

This indicates that the model is still biased toward the majority class in the dataset.

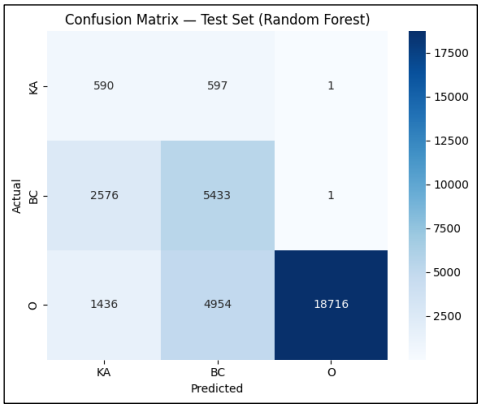


Fig. 2. Confusion matrix of Random Forest

From the Confusion Matrix in Figure 2, it can be seen that the KA class has a high misclassification rate, with 597 out of 1,188 samples incorrectly predicted as the BC class. In contrast, the O class was classified most accurately, with 18,716 correctly predicted samples.

- *LightGBM*

The performance evaluation results of the LightGBM model on the test set are shown in Table VI.

TABLE VI. Accuracy, PRECISION, RECALL, AND F1-SCORE OF LIGHTGBM

Class	precision	recall	f1-score
KA (Severe)	0.1222	0.5101	0.1971
BC (Minor)	0.4952	0.6499	0.5621
O (No Injury)	0.9982	0.7487	0.8557
Accuracy			0.7174
Macro avg	0.5385	0.6363	0.5383
Weighted avg	0.8505	0.7174	0.7643

From Table VI, it can be observed that the LightGBM model achieved an Accuracy of 0.7174 and an F1-macro score of 0.5383. When evaluated by class, the results show a pattern similar to Random Forest. The model performed well in classifying the O (No Injury) class, with an F1-score of 0.8557, while it performed poorly in identifying the KA (Severe) class, with an F1-score of only 0.1971.

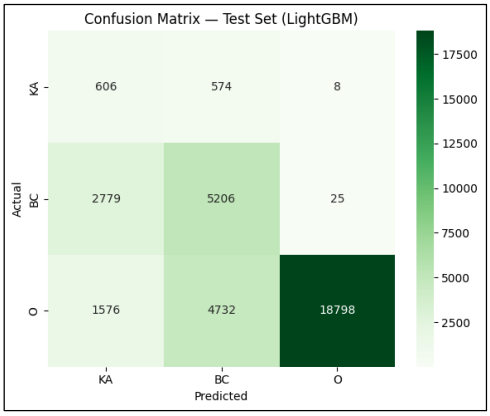


Fig. 3. Confusion matrix of LightGBM

From the Confusion Matrix in Figure 3, it can be seen that the KA class has a high misclassification rate, with 574 samples incorrectly predicted as the BC class and 8 samples incorrectly predicted as the O class, out of a total of 1,188 samples. In contrast, the O class was classified most accurately, with 18,798 correctly predicted samples.

- *CatBoost*

The performance evaluation results of the CatBoost model on the test set are shown in Table VII.

TABLE VII. ACURACY, PRECISION, RECALL, AND F1-SCORE OF CATBOOST

Class	precision	recall	f1-score
KA (Severe)	0.1225	0.5269	0.1988
BC (Minor)	0.4987	0.6427	0.5616
O (No Injury)	0.9981	0.7503	0.8567
Accuracy			0.7174
Macro avg	0.5398	0.6400	0.5390
Weighted avg	0.8512	0.7174	0.7650

From Table VII, it can be observed that the CatBoost model achieved an Accuracy of 0.7174 and an F1-macro score of 0.5390. When evaluated by class, the O (No Injury) class obtained the highest F1-score at 0.8567, while the KA (Severe) class still had the lowest F1-score at 0.1988.

This indicates that even after applying Undersampling, the classification of minority classes remains a significant challenge for the model.

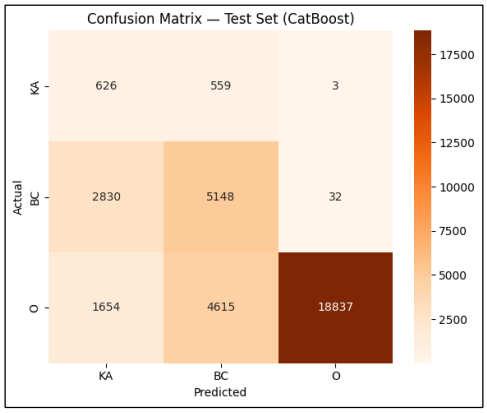


Fig. 4. Confusion Matrix of CatBoost

From the Confusion Matrix in Figure 4, it can be seen that the KA class is frequently misclassified, with 559 samples predicted as the BC class and 3 samples predicted as the O class, out of a total of 1,188 samples. In contrast, the O class was classified most accurately, with 18,837 correctly predicted samples, which is the highest among all individual models.

- *Ensemble*

The performance evaluation results of the Soft Voting Ensemble model on the test set are shown in Table VIII.

TABLE VIII. ACCURACY, PRECISION, RECALL, AND F1-SCORE OF THE SOFT VOTING ENSEMBLE MODEL

Class	precision	recall	f1-score
KA (Severe)	0.1253	0.5210	0.2020
BC (Minor)	0.4961	0.6552	0.5646
O (No Injury)	0.9997	0.7479	0.8557
Accuracy			0.7184
Macro avg	0.5403	0.6414	0.5408
Weighted avg	0.8518	0.7184	0.7651

From Table VIII, it can be observed that the Soft Voting Ensemble model achieved an Accuracy of 0.7184 and an F1-macro score of 0.5408. This performance is very close to the best-performing individual model, Random Forest, which achieved an F1-macro score of 0.5433. This indicates that combining models helps maintain a consistently stable overall performance.

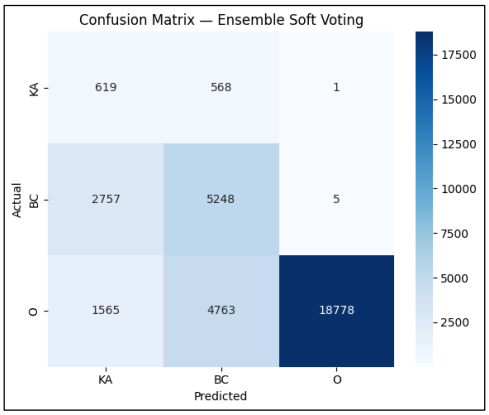


Fig. 5. confusion matrix ของ Ensemble Soft Voting

From the Confusion Matrix in Figure 5, it can be seen that the KA class was correctly classified in 619 samples, which is higher than Random Forest but slightly lower than CatBoost. Meanwhile, the O class was correctly classified in 18,778 samples.

These results are consistent with the characteristics of the dataset, which exhibits a high level of class imbalance—a common limitation faced by all models in this study.

V. CONCLUSION

This study aims to analyze the factors related to the severity of road traffic accidents and develop Machine Learning models for classifying accident severity into three levels: Severe (KA), Minor (BC), and No Injury (O). The computational methods used include Random Forest, LightGBM, CatBoost, and a Soft Voting Ensemble, applied to the Traffic Accidents dataset from Kaggle consisting of 171,519 records.

The performance evaluation results of all models on the test set are summarized in Table IX.

TABLE IX. SUMMARY COMPARISON OF ALL MODELS' PERFORMANCE

Model	Accuracy	Macro Precision	Macro Recall	F1-Macro
RF	0.7212	0.5409	0.6401	0.5433
LightGBM	0.7174	0.5385	0.6363	0.5383
CatBoost	0.7174	0.5398	0.6400	0.5390
Ensemble Soft Voting	0.7184	0.5403	0.6414	0.5408

From Table IX, it can be observed that Random Forest achieved the highest F1-macro score at 0.5433, followed by the

Soft Voting Ensemble at 0.5408. CatBoost and LightGBM obtained F1-macro scores of 0.5390 and 0.5383, respectively.

When focusing on the performance of the KA (Severe) class, which is the most critical category in terms of safety, CatBoost achieved the highest Recall at 0.5269, correctly identifying 626 severe accident cases. This was followed by the Soft Voting Ensemble with a Recall of 0.5210 (619 cases), LightGBM with 0.5101 (606 cases), and Random Forest with the lowest Recall of 0.4966 (590 cases). However, all models exhibited a similar pattern: strong performance in classifying the O (No Injury) class but weak performance in identifying the KA (Severe) class. This reflects that class imbalance remains a major challenge in this study, even after applying Undersampling.

In terms of important features, crash type and first crash type were identified as the most influential variables in Random Forest and CatBoost, while LightGBM placed greater emphasis on crash hour and number of vehicles involved. These differences arise from the distinct learning mechanisms of each algorithm.

VI. Future Directions

To further improve the performance of the model in future studies, two key recommendations are proposed.

First, alternative techniques for handling class imbalance should be applied, such as SMOTE or ADASYN. These methods generate synthetic samples for minority classes instead of removing data from majority classes. This approach may help reduce information loss and improve the model's ability to better classify the KA class.

Second, additional variables related to driver behavior and condition should be incorporated into the model, such as vehicle speed, phone usage while driving, or the driver's physical condition at the time of the accident. These variables are directly related to accident severity and may significantly enhance the model's ability to distinguish high-severity cases more effectively.

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